OTH REGENSBURG

PROJECT REPORT

Cosmin-Andrei Popan

Optimization of an entangled photon source

July 9, 2026

Faculty:	Elektro- und Informationstechnik
Study Programme:	Bachelor Mechatronik
Deadline:	27.07.2026
Supervisor:	Prof. Dr. Ioana Serban
Secondary Supervisor:	Prof. Dr. Robert Sattler

Contents

1 Introduction

This report documents the results of a research project carried out in the Quantum Technology Laboratory of the OTH Regensburg with the aim of optimizing an entangled photon source based on type I spontaneous parametric down-conversion (SPDC) in β-barium borate (BBO) crystals. Firstly, the theory of the SPDC process and the existing laboratory setup are described, highlighting the problems that need to be addressed. Afterwards, a new setup that aims to improve the photon coincidence rates is proposed and implemented, followed by a computer simulation written in Python that compares the measured photon rates with the ones predicted by the theory. Finally, an experiment to measure the Clauser-Horne-Shimony-Holt (CHSH) inequality is performed.

The results show a 20-fold improvement in the coincidence counts, with a CHSH statistic of $S = 2.36 \pm 0.04$, proving the presence of strong entanglement inside the photon pairs.

2 Theoretical Background

Spontaneous parametric down-conversion is a non-linear optical process that converts a photon of higher energy (called a pump photon) into a pair of photons of lower energy (called the signal and idler photons) subject to energy and momentum conservation. This constrant imposed by the conservation laws is referred to as phase-matching. The effect has first been observed in the 1960s and has since become a common technique for developing single-photon sources. One of the possible phase-matching schemes is Type-I SPDC, in which the signal and idler photons that are produced have the same polarization. The necessary conditions for this process can be met by using a birefringent crystal such as β-BBO [1].

2.1 Birefringence in BBO crystals

Birefringence is the optical property of some materials of having an index of refraction that depends on the polarization and direction of light. BBO is a birefringent crystal with a single optical axis. Light propagating through it with polarization perpendicular to its optical axis experiences the so-called ordinary index of refraction n_o and is therefore known as an ordinary ray, whereas light whose polarization lies in the same plane as the optical axis experiences the direction-dependent extraordinary index of refraction $n_e(\theta)$, where θ is the angle between the extraordinary ray and the optical axis. A ray with a different polarization than the two cases described above gets split into an ordinary component and an extraordinary component, each experiencing the respective index of refraction. Denoting with $n_e = n_e(90°)$ the refractive index experienced by the extraordinary ray that propagates perpendicular to the optical axis, the extraordinary refractive index for an arbitrary propagation direction is described by [3]:

$$\frac{1}{n_e^2(\theta)} = \frac{\cos^2(\theta)}{n_o^2} + \frac{\sin^2(\theta)}{n_e^2} \tag{1}$$

The ordinary and principal extraordinary refractive indices are described as functions of the wavelength by the Sellmeier equations [2]:

$$n_o^2 = 2.7359 + \frac{0.01878}{\lambda^2 - 0.01822} - 0.01354\lambda^2 \tag{2}$$

$$n_e^2 = 2.3753 + \frac{0.01224}{\lambda^2 - 0.01667} - 0.01516\lambda^2 \tag{3}$$

2.2 Spontaneous phase-matching in BBO crystals

Type-I spontaneous parametric down-conversion is the process whereby a pump photon of frequency ω_p enters the BBO crystal as an extraordinary ray at an angle θ_m (called the phase-matching angle) with respect to the optical axis and gets converted into signal-idler photon pair with frequencies ω_s, ω_i that exit the crystal as ordinary rays. Therefore, the polarization of the signal and idler rays is perpendicular to that of the pump. Figure 1 depicts the corresponding Feynman diagram. Using $E = \hbar\omega$ and $\mathbf{p} = \hbar\mathbf{k}$, the energy and momentum conservation can be expressed as:

$$\omega_p = \omega_s + \omega_i \tag{4}$$

$$\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i \tag{5}$$

where $\mathbf{k}$ are the respective wave vectors, with the dispersion relation $k = \frac{n\omega}{c}$ [1].

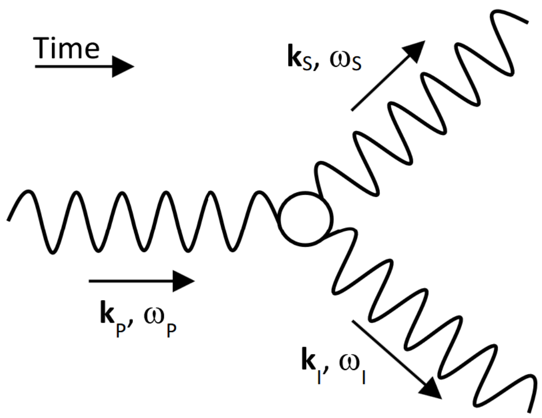

Figure 1: Feynman diagram for SPDC pair production. Figure taken from [4].

Let θ_s and θ_i be the angles formed by the pump ray with the signal and idler rays, respectively (inside the crystal). In the case of degenerate down-conversion, $\omega_s = \omega_i = \frac{\omega_p}{2}$. The momentum conservation in the transverse direction gives $\frac{n_o \omega_s}{c} \cdot \sin(\theta_s) = \frac{n_o \omega_i}{c} \cdot \sin(\theta_i)$, which means $\theta_s = \theta_i$. Longitudinal momentum conservation becomes:

$$n_e(\lambda_p, \theta_m) = n_o(\lambda_s) \cdot \cos(\theta_s) \tag{6}$$

Therefore, the photons are symmetric?? something about refraction at exit

2.3 SPDC intensity

2.4 Temperature dependence of laser wavelength

2.5 Photon entanglement and the CHSH inequality

List of Figures

References

[1] Feng-Kuo Hsu and Chih-Wei Lai. Absolute instrument spectral response measurements using angle-resolved parametric fluorescence. *Optics Express*, 21(15):18538, July 2013.

[2] K. Kato. Second-harmonic generation to 2048 Å in b-ba2o4. *IEEE Journal of Quantum Electronics*, 22(7):1013–1014, 1986.

[3] B.E.A. Saleh and M.C. Teich. *Polarization and Crystal Optics*, chapter 6, pages 193–237. John Wiley and Sons, Ltd, 1991.

[4] Neil A. Salmon and Stephen R. Hoon. A millimeter-wave bell test using a ferrite parametric amplifier and a homodyne interferometer. *Journal of Magnetism and Magnetic Materials*, 501:166435, May 2020.

Erklärung

1. Mir ist bekannt, dass dieses Exemplar der Project Report als Prüfungsleistung in das Eigentum der Ostbayerischen Technischen Hochschule Regensburg übergeht.
2. Ich erkläre hiermit, dass ich diese Project Report selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet habe.

Ort, Datum und Unterschrift

Presented by:	Cosmin-Andrei Popan
Student ID:	3434442
Study Programme:	Bachelor Mechatronik
Time Frame:	16.03.2026 – 27.07.2026
Supervisor:	Prof. Dr. Ioana Serban
Secondary Supervisor:	Prof. Dr. Robert Sattler